A NEXT-GENERATION AI FRAMEWORK

FOR INTELLIGENT TRAVEL SUPPORT FOR SRI LANKA

R26-IT-112

AI Travel Analyzer | Emotion Recognition | Transport Verification | Immigration Compliance

SLIIT · Faculty of Computing · Department of Information Technology · February 2026

Project Introduction

An integrated AI platform combining four components — travel planning, emotion recognition, transport safety, and immigration compliance — to make Sri Lanka tourism safer, smarter, and legally compliant for foreign tourists.

Key Problems

- No immigration or compliance assistance.
- Reactive systems — respond only after user queries.
- No optimized travel package generation.
- No real-time monitoring of disruptions.
- Limited tourist safety & transport verification.

Social & Research Impact

- Reduces visa violations & supports 300,000+ jobs.
- AI-driven compliance, verified safety & disruption handling.
- First platform combining all 4 tourism AI modules.

Previous Research & Our Solution

Research Gaps

- Prior systems address planning, compliance, safety & emotion in isolation, not together.
- Existing travel platforms are reactive rather than proactive.
- No integrated framework combining all four dimensions.
- Real-time proactive tourist guidance unaddressed in Sri Lanka.

Our Solution

A next-generation AI framework that proactively integrates travel planning, immigration compliance, emotion recognition, and transport safety in one unified platform for Sri Lanka tourism.

Constraints & Limitations

- Data: Real tourist profiles & live immigration regulatory data
- Privacy: Biometric, facial & personal travel data handling
- Infrastructure: Connectivity gaps in rural Sri Lanka
- Adoption: Resistance from tourists & operators

Our Integrated Solution

A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka — four integrated components:

1 AI Conversational Travel Analyzer

(Component 1)

Generates optimized Sri Lanka itineraries and proactively re-routes tourists before disruptions occur.

2 Adaptive Emotion Recognition

(Component 2)

Identifies areas of low satisfaction, provides actionable recommendations and supports business development by increasing visitor satisfaction.

3 Smart Safety & Transport Verification

(Component 3)

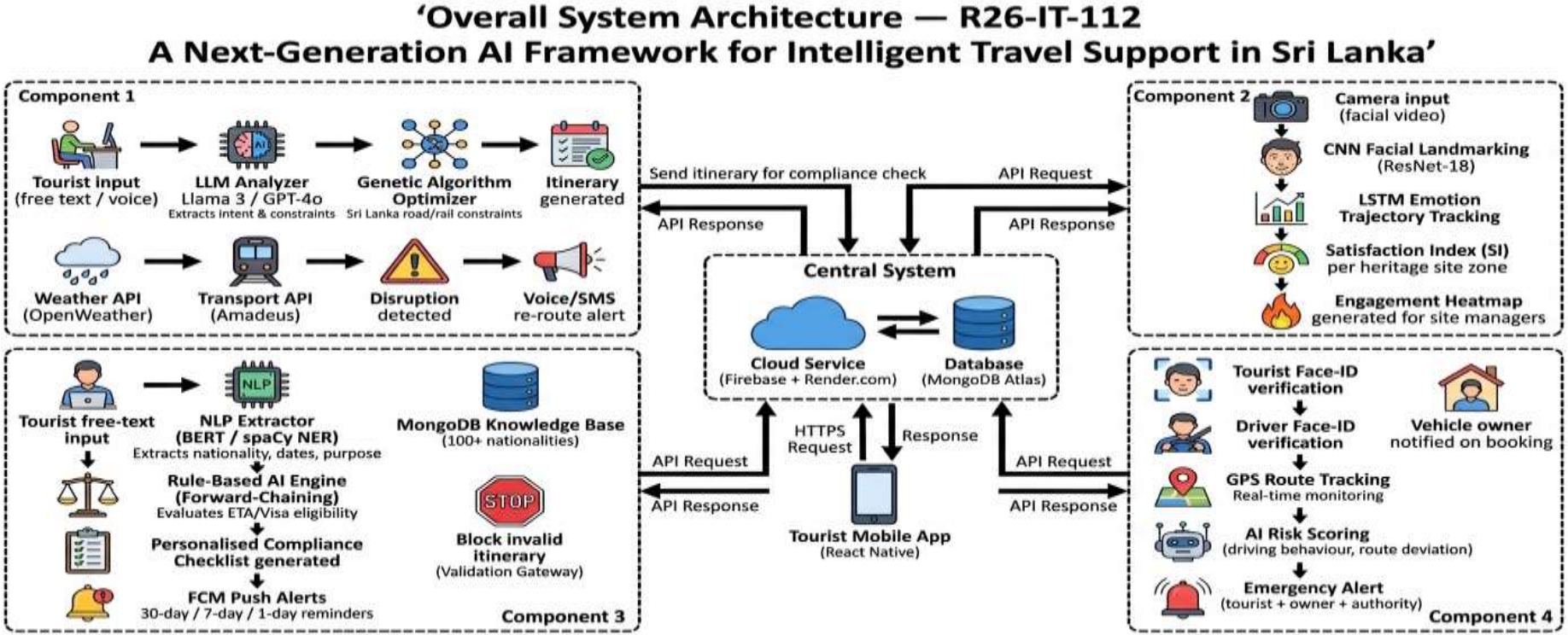
Verifies tourist and driver identity using dual Face-ID with continuous AI-based risk scoring.

4 AI-Based Immigration & Travel Compliance

(Component 4)

Evaluates visa eligibility, generates personalized compliance checklists, and delivers proactive deadline alerts.

Overall System Diagram





IT22082442 | SENEVIRATHNA T.M.N.D.L

AI Conversational Travel Analyzer

with Intelligent Package Generation and Real-Time Adaptive Engine

AI Conversational Travel Analyzer

with Intelligent Package Generation and Real-Time Adaptive Engine

Contribution:

Provides the complete AI travel intelligence layer — multi-intent NLP extraction, Genetic Algorithm package optimization, and proactive real-time adaptive itinerary management.

Novelty:

First system combining Multi-Intent NLP + Genetic Algorithm package optimization + Real-Time Guardian Loop + Proactive Voice Agent specifically for Sri Lanka travel.

Novelty & Problem Definition

Core Problem: No existing system simultaneously handles multi-intent travel requests, optimizes packages with Genetic Algorithms, AND proactively adapts in real time to Sri Lanka-specific disruptions.

Gap 1	No platform extracting multi-intent travel constraints via LLM	**Multi-Intent LLM:** Llama-3/GPT-4o extracts budget, destinations, duration, transport style simultaneously.
Gap 2	No Genetic Algorithm travel package optimization for Sri Lanka	**Genetic Algorithm:** Pareto-optimized package balancing cost, time, and comfort within LKR 4 lakh budget.
Gap 3	No terrain/speed-aware travel time modeling for local roads	**Guardian Loop:** Continuous real-time monitoring of weather, flights, traffic and crowd APIs.
Gap 4	No real-time Guardian Loop monitoring weather + traffic + flights	**Proactive Voice:** Auto-notifies user with updated itinerary before disruption impacts the trip.

Domain Knowledge & Research Objectives

MAIN OBJECTIVE: To design and evaluate an AI-based proactive travel planning system that generates optimized personalized vacation packages and automatically adapts them based on real-time environmental and transport disruptions.

SO1 Multi-Intent NLP Analyzer using LLMs (Llama-3/GPT-4o) — extract budget, destinations, duration and transport preferences from free text.

SO2 Genetic Algorithm optimizing packages within LKR 4 lakh — balancing flights, hotels, transport, comfort and time.

SO3 Sri Lanka-specific travel time model — terrain, road types and speed variations.

SO4 Real-Time Guardian Loop — monitors weather, flight status, traffic and crowd data continuously.

SO5 Proactive Voice Agent — notifies users and suggests alternatives before disruptions hit.

SO6 Evaluate: NLP accuracy, GA convergence, budget efficiency, response time <3s.

Methodology & Technology Stack



Research Approach

Design Science Research (DSR)
Experimental validation
Benchmarked on 100 simulated scenarios
Comparison with existing platforms



AI & ML Techniques

Llama-3 / GPT-4o LLM — NLP extraction
Genetic Algorithm — package optimization
Pareto multi-objective optimization
Real-time risk score calculation



Tools & Platforms

Python (FastAPI backend)
React Native (mobile app)
Weather / Flight / Traffic APIs
PostgreSQL + cloud microservices



Validation & Ethics

NLP accuracy, GA convergence rate
Budget efficiency, response time <3s
No personal data without consent
PDPA Sri Lanka compliance

Implementation Details & System Architecture

01

LAYER 1: User Input (NLP Analyzer)

Natural language request via mobile app · LLM extracts: budget, destinations, duration, transport style · Converts unstructured text → structured constraints

02

LAYER 2: Package Generation (Genetic Algorithm)

Pareto multi-objective optimization · Selects flights, hotels, transport within ≤ LKR 4L · Accounts for Sri Lanka terrain & travel time variations

03

LAYER 3: Real-Time Adaptive Engine

Guardian Loop: Weather + Flight + Traffic APIs · Risk score on disruption breach · Re-runs GA, updates itinerary automatically

04

LAYER 4: Proactive Voice Output

Voice Agent alerts BEFORE disruption impacts trip · Suggests alternative destinations or updated plans · Mobile push notifications + voice-first UX

DATA FLOW: User Input → LLM NLP → Constraints → Genetic Algorithm → Travel Package → Guardian Loop → Risk Score → Re-Optimize → Voice Notification

Dependencies & Common Components: GPS Module · Central AI Platform API · Tourist Mobile App (React Native) · Weather/Flight/Traffic APIs



IT22216014 | KAWEESHWARA M.D.O

Adaptive Emotion Recognition

for Tourist Experience Management

Adaptive Emotion Recognition

for Tourist Experience Management

Contribution:

Provides full emotional intelligence layer for the AI Travel Framework - real-time emotion data, zone satisfaction mapping, and recommendation outputs.

Novelty:

First mobile app combining multi-modal emotion recognition with zone mapping , emotional tracking overtime , anomaly detection. The system identifies areas of low satisfaction, provides actionable recommendations for supports business development by optimizing operations and increasing visitor satisfaction.

Novelty & Problem Definition

Core Problem: Tourist destinations cannot detect how visitors feel at specific locations in real time -making it impossible to improve experiences before they are already over.

Gap 1	No mobile app platform for tourist emotion recognition	**Mobile-First Platform:** Zero hardware. Any smartphone, any site. Scales to 1000s of simultaneous tourists.
Gap 2	No zone-level spatial emotion mapping	**Zone-Level Emotion Mapping:** Links emotion to exact zone — 'Exhibit B causes boredom at 2pm'. First in tourism literature.
Gap 3	No tracking of emotional changes over time	**Emotional Trajectory:** LSTM tracks how emotions evolve across zones - transforms snapshots into visitor journey narratives.
Gap 4	No real-time anomaly detection for visitor problems	**Anomaly Detection:** **Flags sustained distress/boredom triggers staff & manager alerts automatically.**
Gap 5	No closed-loop system linking emotion data to business improvements	**Business Optimization:** Actionable recommendations to improve operations and increase visitor satisfaction.

Domain Knowledge & Research Objectives

MAIN OBJECTIVE: To design, develop, and validate a real-time mobile Adaptive Emotion Recognition system — multi-modal data, zone mapping, trajectory tracking, anomaly detection, and actionable recommendations for tourist experience improvement.

RESEARCH OBJECTIVES

- **SO1** Facial emotion recognition module using MobileNetV2 (FER2013)
- **SO2** Multi-modal fusion combining facial, location, crowd density and time.
- **SO3** Build LSTM Emotional Trajectory tracker - emotional patterns across zones.
- **SO4** Anomaly detection - flag distress/boredom, trigger staff/manager alerts.
- **SO5** Destination manager dashboard - heatmaps, satisfaction trends, AI recommendations.

TECHNOLOGIES

AI & ML:

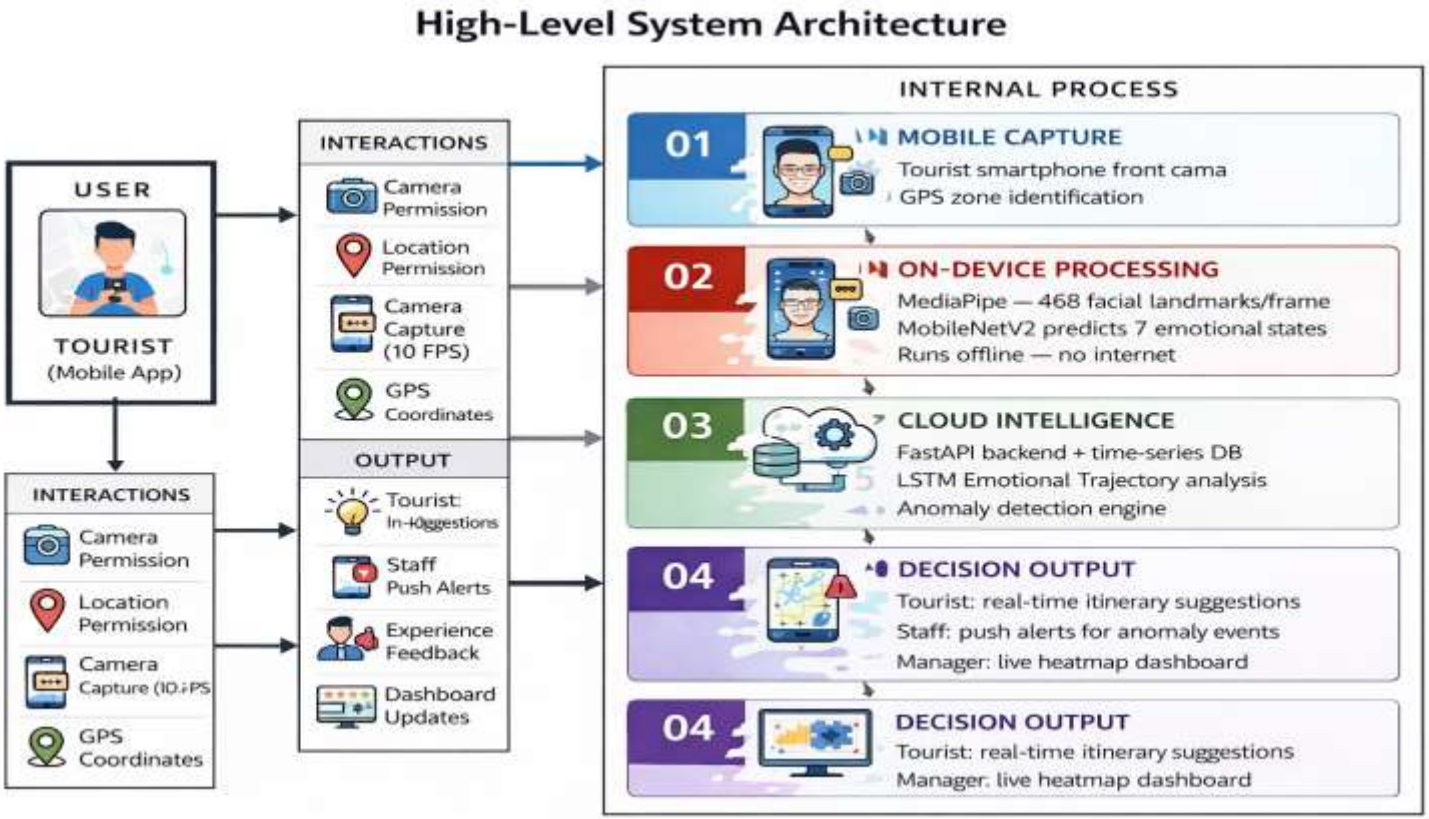
MobileNetV2 CNN · MediaPipe Face Mesh (468 landmarks) · LSTM temporal trajectory modeling · Late-fusion multi-modal pipeline

Mobile & Backend:

React Native (Android + iOS) · TensorFlow Lite (on-device) · FastAPI backend · YOLO v8 crowd density

Validation & Ethics:

Target: >80% accuracy, <150ms latency , Informed consent · No raw images -anonymized vectors only


High-Level System Architecture
USER
TOURIST
(Mobile App)
INTERACTIONS
Camera Permission
Location Permission
Camera Capture (10 FPS)
GPS Coordinates
OUTPUT
Tourist: In-ride suggestions
Staff Push Alerts
Experience Feedback
Dashboard Updates
INTERACTIONS
Camera Permission
Location Permission
Camera Capture (10 FPS)
GPS Coordinates
INTERNAL PROCESS
01
MOBILE CAPTURE
Tourist smartphone front cama
GPS zone identification
02
ON-DEVICE PROCESSING
MediaPipe — 468 facial landmarks/frame
MobileNetV2 predicts 7 emotional states
Runs offline — no internet
03
CLOUD INTELLIGENCE
FastAPI backend + time-series DB
LSTM Emotional Trajectory analysis
Anomaly detection engine
04
DECISION OUTPUT
Tourist: real-time itinerary suggestions
Staff: push alerts for anomaly events
Manager: live heatmap dashboard
04
DECISION OUTPUT
Tourist: real-time itinerary suggestions
Manager: live heatmap dashboard



I T 2 2 1 7 6 9 7 4 | R U P A S I N G H E
M . L . K . S

Smart Safety & Transport Verification System

An AI-Powered Tourist Transport Safety Framework for Sri Lanka

Smart Safety & Transport Verification System

Student Photo

An AI-Powered Tourist Transport Safety Framework for Sri Lanka

Contribution:

Provides the full transport safety layer for the Smart Tourism Ecosystem — verified identities, real-time GPS monitoring, AI risk scoring, and emergency alert outputs.

Novelty:

First mobile platform combining CNN facial verification + GPS anomaly detection + XGBoost risk scoring + proactive emergency alerting for tourist transport in Sri Lanka.

Novelty & Problem Definition

Core Problem: Sri Lanka lacks a fully integrated digital platform for tourist transport that combines identity verification with real-time AI safety monitoring leaving tourists vulnerable during ground travel.

Gap 1	No facial identity verification for tourist transport boarding	**Face ID:** we are create Mobile CNN base verification system**.using MTCNN,Face net**
Gap 2	No GPS monitoring linked to identity confidence or risk scoring	**GPS + Deviation Alerts:** we create real time **GPS** Tracking system With risk analys
Gap 3	Emergency systems are reactive no proactive pre-incident alerting	**XGBoost Risk Scoring:** we are creat **XGBOOST** based Risk Scoring
Gap 4	No automatic vehicle owner notification mechanism	**Proactive Emergency System:** Predicts risk before incidents alerts tourist, owner, and emergency services in real-time.
Gap 5	No cumulative safety record integrated with travel itineraries	**Unified Safety Record:** Full journey safety timeline accessible to tourists, operators and emergency services.

Domain Knowledge & Research Objectives

MAIN OBJECTIVE: To design, develop, and validate a Smart Safety & Transport Verification System integrating facial recognition, GPS monitoring, AI-based risk scoring, and real-time emergency alerting for verified, safe, and accountable tourist transport in Sri Lanka.

SO1 Create a Face-ID system (MTCNN + FaceNet) to check the tourist and driver identity at boarding with at least 90% accuracy.

SO2 Set up real-time GPS monitoring to spot route deviations over 500m and unusual speeds, sending alerts within 30 seconds.

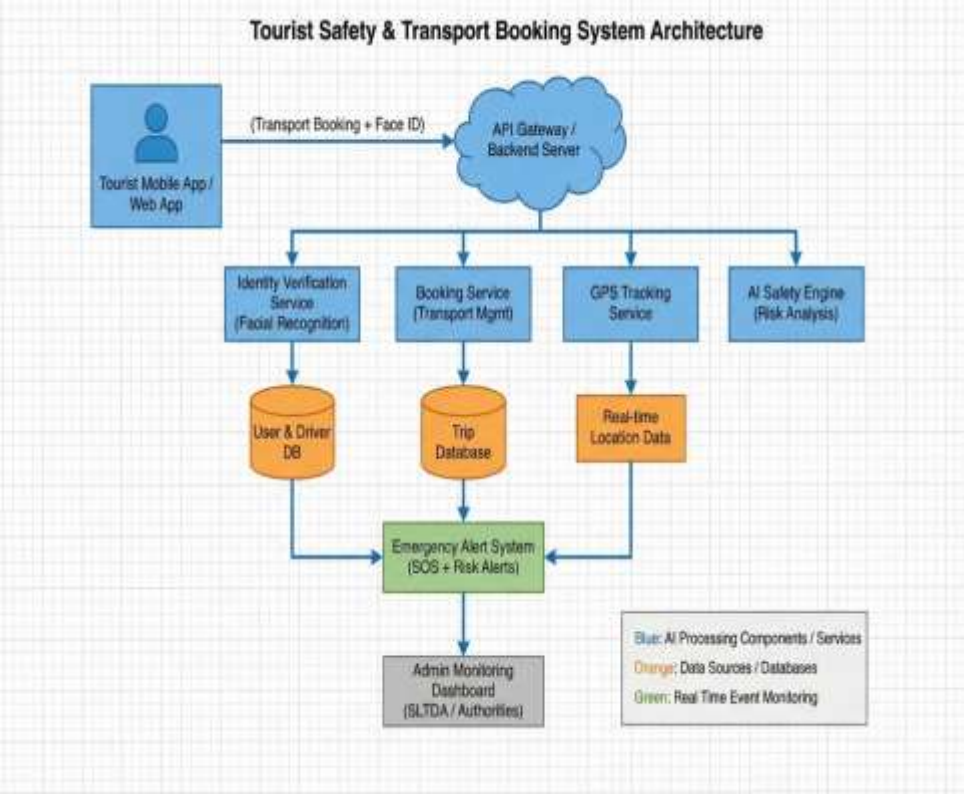
SO3 Build an XGBoost risk scoring model using 47 features from driver profile, journey, and telemetry, aiming for over 91% accuracy

SO4 Automated owner notification push alerts within 10 seconds of booking or safety events.

SO5 Integrate with Smart Tourism Ecosystem unified safety timeline accessible to all parties.

SO6 Test the system with at least 50 journeys to check face recognition accuracy, GPS alert speed, risk model performance, and usability.

ADAPTIVE EMOTION RECOGNITION | Implementation Details



01 **LAYER 1: Mobile Capture**

The phone camera captures the **user** and the GPS tracks the **location**."

02 **LAYER 2: On-Device Processing**

The phone checks the face and makes sure it is a **real person**, not a photo

03 **LAYER 3: Cloud Intelligence**

The phone checks the face and makes sure it is a **real person**, not a photo."

04 **LAYER 4: Decision Output**

"Finally, the **Tourist** gets safety alerts and the **Manager** can monitor everything live."



IT22607614 | MANTHRIRATHNA M.A.A.C

AI-Based Immigration & Travel Compliance

Automated Visa Eligibility, Compliance Checklists & Proactive Alerts

AI Based Immigration & Travel Compliance

Contribution:

Developed a system that checks tourist travel details and automatically decides visa eligibility using Sri Lanka immigration rules.

Novelty:

The novelty of this research lies in **transforming complex visa regulations into an automated decision-support system** that can evaluate travel plans and provide real-time visa compliance guidance for tourists visiting Sri Lanka.

Manthrirathne M.A.A.C · IT22607614

A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka

NOVELTY & PROBLEM DEFINITION

RESEARCH PROBLEM & GAP

Core Problem: Existing travel platforms rely on static visa portals — tourists manually navigate complex ETA rules, leading to overstays and legally infeasible itineraries confirmed without detection.

- **Gap 1:** Tourists receive no automatic reminders about visa deadlines or missing documents.
- **Gap 2:** No system generates a personalized document checklist based on the tourist's nationality.
- **Gap 3:** Visa and ETA eligibility is never checked in real time during the booking process.

NOVELTY

NLP Extraction Engine: BERT/spaCy NER extracts nationality, dates, purpose from free-text — no manual input required.

Rule-Based AI Compliance: Forward-chaining inference engine evaluates Sri Lanka ETA and visa rules for 100+ nationalities.

Itinerary Validator: Blocks legally infeasible travel plans before booking confirmation — prevents overstays.

Proactive FCM Alerts: Firebase Cloud Messaging delivers 30-day, 7-day, and 1-day deadline reminders automatically.

Personalized Checklists: Generates tailored visa/ETA checklist per tourist nationality stored in versioned MongoDB.

DOMAIN KNOWLEDGE & RESEARCH OBJECTIVES

MAIN OBJECTIVE: To develop an AI system that checks visa eligibility, generates compliance checklists, and delivers deadline alerts integrated into travel booking confirmation.

RESEARCH OBJECTIVES

- **SO1** NLP pipeline (BERT/spaCy) — extract nationality, dates, purpose from free text.
- **SO2** Rule-based AI inference engine — Sri Lanka visa/ETA rules for 100+ nationalities.
- **SO3** Itinerary validation — block legally infeasible trips before booking.
- **SO4** Personalized compliance checklist per nationality in versioned MongoDB.
- **SO5** FCM push notifications — proactive deadline alerts at 30, 7, and 1 day.
- **SO6** Validate: ≥90% accuracy on 200+ nationality scenarios, API ≤2 seconds.

LATEST TECHNOLOGIES USED

NLP & AI:

BERT/spaCy NER · Forward-chaining inference engine · Google Colab Pro (GPU-accelerated NLP fine-tuning)

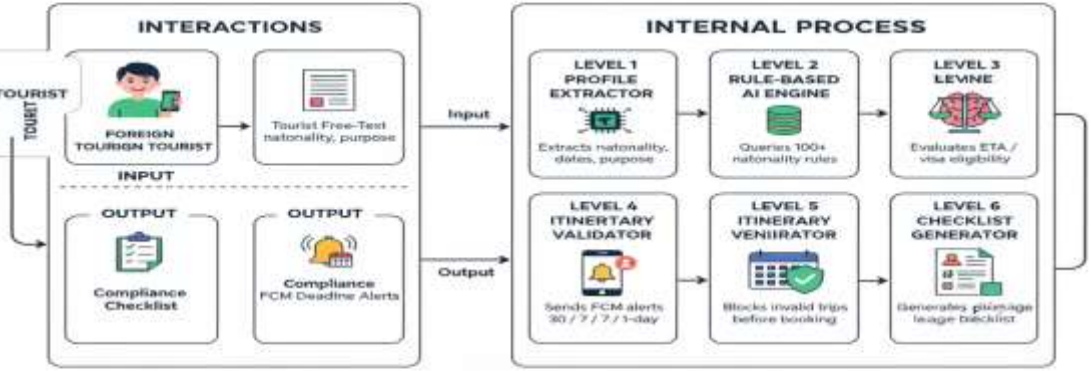
Backend & Database:

Node.js · Express.js · Python Flask · MongoDB Atlas (versioned schema) · Firebase Cloud Messaging (FCM)

Validation & Ethics:

≥90% compliance accuracy · API ≤2s · All data anonymized · PDPA Sri Lanka Act No. 9 of 2022

IMPLEMENTATION DETAILS & SYSTEM ARCHITECTURE



Dependencies & Integration

- NLP Extractor → Rule Engine → MongoDB Knowledge Base → Checklist Generator → Firebase FCM Alert Engine.
- Itinerary Validation Gateway intercepts booking confirmation : blocks legally infeasible trips before confirmation

COMMERCIALIZATION

Investment & Recovery

- Total group budget: LKR 236,225 (open-source stack — zero licensing cost).
- Recovery: B2B SaaS (LKR 15,000–25,000/month per operator), B2G licensing with SLTDA, premium consumer model (LKR 500/trip premium).

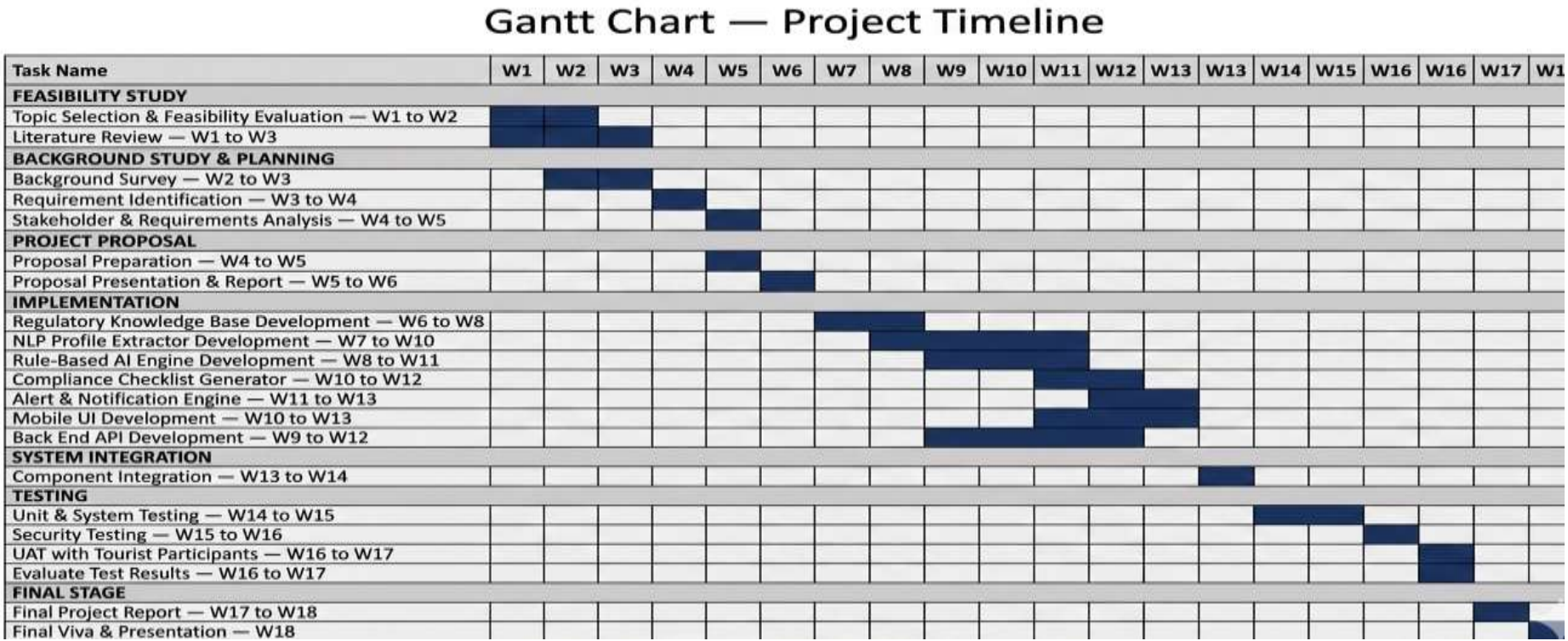
Target Market

- Only platform combining travel planning + compliance + emotion recognition + transport safety.

Intellectual Property Rights

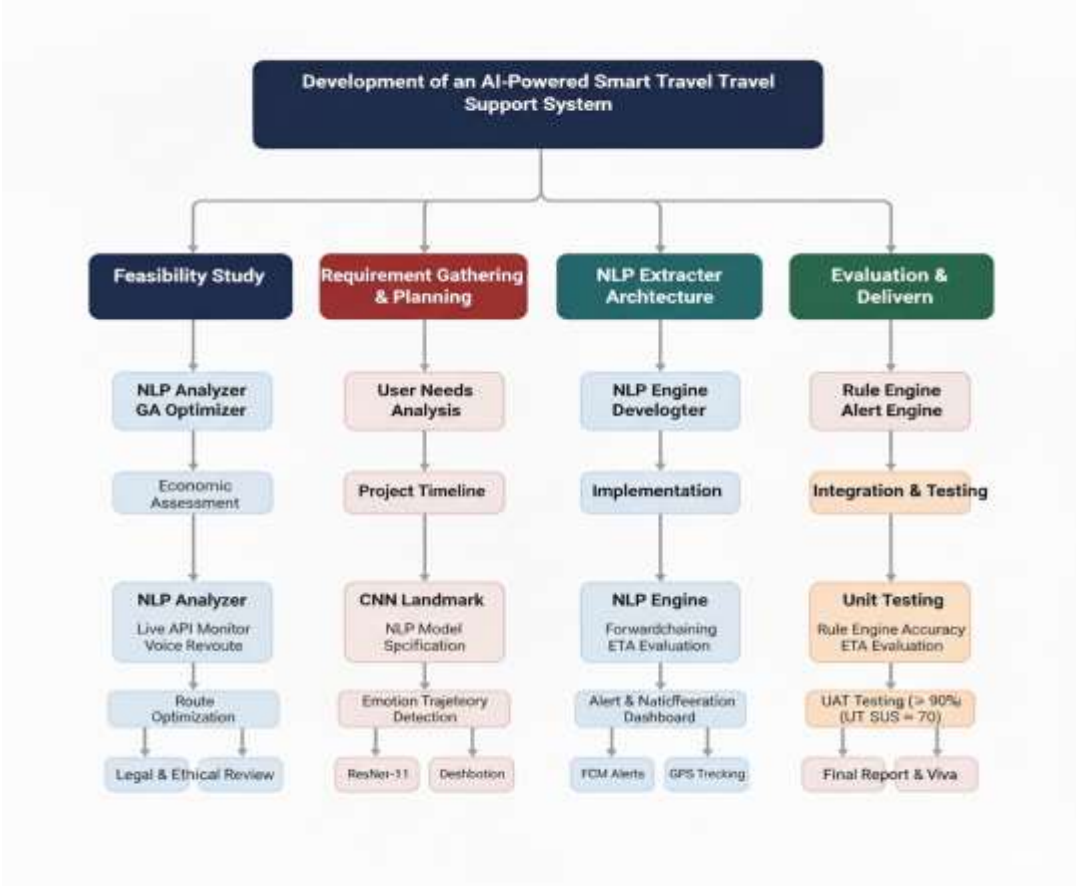
- IP owned by SLIIT research team under academic policy.
- Patent potential for the Itinerary Validation Gateway integration method

Grant Chart


Gantt Chart — Project Timeline
Task Name
W1
W2
W3
W4
W5
W6
W7
W8
W9
W10
W11
W12
W13
W13
W14
W15
W16
W16
W17
W1
FEASIBILITY STUDY
Topic Selection & Feasibility Evaluation — W1 to W2
Literature Review — W1 to W3
BACKGROUND STUDY & PLANNING
Background Survey — W2 to W3
Requirement Identification — W3 to W4
Stakeholder & Requirements Analysis — W4 to W5
PROJECT PROPOSAL
Proposal Preparation — W4 to W5
Proposal Presentation & Report — W5 to W6
IMPLEMENTATION
Regulatory Knowledge Base Development — W6 to W8
NLP Profile Extractor Development — W7 to W10
Rule-Based AI Engine Development — W8 to W11
Compliance Checklist Generator — W10 to W12
Alert & Notification Engine — W11 to W13
Mobile UI Development — W10 to W13
Back End API Development — W9 to W12
SYSTEM INTEGRATION
Component Integration — W13 to W14
TESTING
Unit & System Testing — W14 to W15
Security Testing — W15 to W16
UAT with Tourist Participants — W16 to W17
Evaluate Test Results — W16 to W17
FINAL STAGE
Final Project Report — W17 to W18
Final Viva & Presentation — W18

Grant Chart| Diagram

Work Breakdown Structure



Thank You

A Next-Generation AI Framework for Intelligent Travel Support for Sri Lanka

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